Question		Answer	Marks	AO	Guidance
10	(a)	A particle is an object [with mass] with negligible size	B1 [1]	1.2	Also allow object whose size and shape does not matter. Accept point mass. Do not award if other wrong statements also made.
10	(b)	Weight is $-3g\mathbf{j}$ For equilibrium $-3g\mathbf{j} + h\mathbf{i} + k\mathbf{i} + (k-5)\mathbf{j} = \mathbf{0}$ In the **j** direction $-3g + k - 5 = 0$ [so $k = 34.4$] **i** direction $h + k = 0$ so $h = -34.4$	B1 M1 A1 [3]	1.1 3.1b 1.1	Accept $-29.4\mathbf{j}$. May be implied by the use of weight (with correct sign) in their vertical equation and not their horizontal equation. May be implied by $k = 34.4$ seen with no equation Vector equation need not be seen explicitly Uses the **j**-direction to find a value for k. Allow only sign errors. Need not be simplified. May be implied by $k = 34.4$ seen Cao
10	(c)	$h\mathbf{i} - 3g\mathbf{j} = 3\mathbf{a}$	B1FT [1]	2.5	Must be vector equation FT their h from (b) and accept numerical equivalents (e.g. note $-34.4\mathbf{i} - 29.4\mathbf{j} = 3\mathbf{a}$ is fully correct) ISW if **a** is evaluated